

Divyanshu Kumar

AI & Robotics Engineer

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LinkedIn — GitHub — Portfolio

PROFESSIONAL SUMMARY

AI & Robotics Engineer with experience building intelligent embedded systems integrating computer vision, IoT sensors, and real-time AI inference. Skilled in developing edge AI pipelines, robotics control logic, and backend services for hardware-software integration. Strong background in deep learning, system architecture, and deployment on Linux-based embedded platforms.

EDUCATION

B.Tech in Computer Science Engineering

Dumka Engineering College, 2025

CGPA: 8.55/10

EXPERIENCE

Tech Genius Institute

Aug 2025 – Present

AI & Robotics Engineer / Mentor

- Designed intelligent robotics systems integrating sensors, microcontrollers, and AI inference pipelines for real-time decision making.
- Developed Python-based control services enabling communication between embedded devices and AI modules using structured APIs and messaging.
- Built human-robot interaction prototypes using speech synthesis, LLM-based reasoning, and multimodal inputs on Linux edge devices.
- Led end-to-end project lifecycle including architecture design, development, testing, and deployment-ready prototypes.

Gyantu (Remote)

Nov 2024 – Jan 2025

AI Intern

- Developed modular AI backend for conversational systems using LLaMA 3.2, LangChain, and ChromaDB.
- Implemented Retrieval-Augmented Generation pipelines including document ingestion, embedding creation, and semantic search.
- Designed agent-based workflows for intelligent query routing and response generation.
- Built Streamlit interface with clear backend-frontend separation improving response quality by 40%.

PROJECTS

AI-Powered Driver Assistance & Monitoring System

- Developed real-time edge AI system integrating computer vision models for drowsiness detection, emotion recognition, and driver behavior analysis.
- Implemented CNN and 3D-CNN inference pipelines with Flask-based backend enabling live video processing and alert generation.
- Integrated IoT hardware (NodeMCU, ultrasonic sensors, IR sensors) using MQTT communication for vehicle feedback and safety alerts.
- Designed complete pipeline from sensor acquisition to intelligent decision logic and actuator response.

Interactive Talking Robot (LLM + Voice AI on Raspberry Pi 5)

- Built voice-interactive robotic assistant running on Raspberry Pi 5 with Linux OS integrating LLM reasoning and ElevenLabs speech synthesis.
- Implemented speech input processing, conversational logic, and audio response generation for natural human-robot interaction.
- Developed modular architecture enabling hardware control triggers based on conversational context.
- Optimized inference and audio pipeline for low-latency performance on edge hardware.

Gene Ontology Classification using Deep Learning

- Engineered preprocessing pipeline for high-dimensional biological datasets with feature filtering and encoding.
- Implemented TabTransformer-based models achieving improved classification performance.
- Explored Positive-Unlabeled learning strategies to handle severe class imbalance.

TECHNICAL SKILLS

Languages: Python, SQL, C++

AI/ML: Deep Learning, Computer Vision, Transformers, LLMs, RAG

Robotics & Embedded: Raspberry Pi, NodeMCU/ESP8266, Sensors (Ultrasonic, IR), Edge AI

Backend: Flask, REST APIs, System Architecture, Modular Design

Messaging: MQTT, Real-Time Communication

Tools: Linux, Git, Docker (Basics), CI/CD Concepts, AWS/GCP Fundamentals

ACHIEVEMENTS

Winner — Innovathon (BIT Sindri): Smart Grocery Quality Control System

Finalist — HackFest '24 (IIT ISM): Unified Customer Insights

Best Final Year Project — AI Driver Assistance System

Solved 250+ Data Structures & Algorithms Problems